CHAPTER 2

The Contemporary Standardized Survey Interview for Social Research*

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2.1 INTRODUCTION

Investigators in many fields have long relied on research interviews conducted as part of sample surveys as a major source of data. This chapter introduces readers to the justification for the standardized survey interview, summarizes the rules that standardized interviewers follow, and describes some features of interaction in the interview. We briefly enumerate some actions in which interviewers and respondents engage and describe how the respondent's answer sometimes displays conversational practices learned in other social situations. We then present a detailed analysis of reports, a type of answer to a survey question that is not in the format projected by the question, which illustrates one mechanism by which respondents display problems

*Some of the research reported here was supported in part by the National Institute of Child Health and Human Development (Center Grant R24 HD047873), by the National Institute on Aging (Center Grant P30 AG017266 and the Wisconsin Longitudinal Study: Tracking the Life Course P01 AG021079), by the University of Wisconsin Center for Demography and Ecology, by the University of Wisconsin Center for Demography of Health and Aging, by grants from the Graduate School Research Committee to Schaeffer and Maynard, and by the University of Wisconsin Survey Center (UWSC). Nora Cate Schaeffer and Douglas W. Maynard are coinvestigators on the WLS Supplement "Cognition and Interaction in Surveys of Older Adults." Collaborators are Jennifer Dykema (UWSC), Dana Garbarski (Department of Sociology), and Cabell Gathman (Department of Sociology).

Envisioning the Survey Interview of the Future, Edited by Frederick G. Conrad and Michael F. Schober

in answering survey questions.[1] In our discussion we consider some implications of such studies of interaction for survey interviews in the future.

2.2 THE SURVEY RESEARCH INTERVIEW

The goal of most social research surveys is to describe a population. Like other research goals this one entails certain requirements. One requirement is that units of observation must be selected with known, nonzero probabilities of selection; this is part of what makes the data collection effort a "survey." In addition, the data collected from the units in the sample, the respondents, must be comparable. For the data to be logically comparable across units, all respondents must be asked the same questions (with the same meaning across respondents), offered the same exhaustive and mutually exclusive categories, and answer all questions. The rationale for these logical conditions is straightforward: if, for example, respondents are asked different questions (or questions with different meanings), then variation in their answers could be due to variation in the wording of the question, not to differences in the underlying concept being measured. If one interviewer asks, "Do you favor or oppose Governor Doyle's veto of this bill?" and another interviewer asks, "Do you favor Governor Doyle's veto of this bill?" the answers are not logically comparable. If all respondents are not given the same opportunities to provide each item of information, then the absence of a response becomes difficult to interpret: Did the question not apply, or was it not asked?

Survey researchers think of the questions asked in a survey interview as measurement tools, and survey interview protocols are commonly referred to as "instruments." Survey methodologists use two criteria derived from measurement theory to evaluate the quality of survey questions. The criterion of validity addresses whether the question measures the concept it is intended to measure. The criterion of reliability addresses whether a question would be answered in the same way by other respondents with the same true value (or by the same respondent if she or he were asked again and the underlying true value had not changed). Because assessing the reliability or validity of a survey item is itself a substantial undertaking, researchers often rely on surrogate assessments—indicators that are known or presumed to be associated with validity or reliability—to evaluate survey questions. These indicators include, for example, the results of cognitive interviews, a method of investigating cognitive processing that has been adapted for testing survey questions, in which subjects think out loud as they answer the survey questions (see Presser et al., 2004 for examples). Similarly, researchers have coded the interaction between the interviewer and the respondent to see whether the behaviors of the actors suggests that the questions cannot

[1] "Reports" refer to a class of answers that do not match the format projected by a question. For example, if a question projects a "yes" or "no" answer (e.g., "Can you read ordinary newsprint without glasses?"), and the respondent gives an answer other than "yes" or "no" (e.g., "Only reading glasses"), the answer does not match the format projected by the question. We use "reports" to refer to this type of answer, and not to "self-reports" more generally. (See Moore, 2004; Schaeffer and Maynard, 1995).

be administered in a standardized way or that the answers may have measurement error (e.g., see Draisma and Dijkstra, 2004; Schaeffer and Dykema, 2004). Measurement error may be invalidity (measuring something other than the target concept) or unreliability (variability in the answers that is due to something other than the concept being measured).

Because questions in surveys are instruments for measurement, survey researchers attempt to improve their validity and reliability in a variety of ways. For example, aids such as calendars or questions that provide memory cues may be used to improve the accuracy of answers. Researchers also attempt to write questions that will be interpreted consistently by respondents and thus provide more reliable data. Because surveys often use many interviewers to collect data from thousands of respondents, researchers have long been concerned with the effect different interviewers might have on the answers that are recorded (Hyman, 1954/1975). The practices of standardization have been developed to attempt to hold the effect of the interviewer constant and thereby reduce unreliability due to variation in the behavior of interviewers across respondents.

2.3 STANDARDIZATION IN THE SURVEY INTERVIEW[2]

Standardization refers to a set of practices that have developed over many years, and that are defined and implemented in different ways at different institutions (Viterna and Maynard, 2002). In *Standardized Survey Interviewing*, Fowler and Mangione (1990) codified a comprehensive set of rules that were grounded in the rationale for standardization and refined through many years of practice and observation. Fowler and Mangione (1990, p. 35) describe four principles of standardized interviewing:

1. Read questions as written.
2. Probe inadequate answers nondirectively.
3. Record answers without discretion.
4. Be interpersonally nonjudgmental regarding the substance of answers.

Although survey researchers would probably universally accept these principles, researchers and staff charged with supervising field operations at various centers have, over the years, developed their own specifications for training interviewers, sometimes importing practices from other centers (Viterna and Maynard, 2002, p. 393). Although the 12 academic survey centers that Viterna and Maynard examine vary considerably in their practices, “all centers in this study require all questions

[2]This section is based on Douglas W. Maynard and Nora Cate Schaeffer (2002), standardization and its discontents. In *Standardization and Tacit Knowledge: Interaction and Practice in the Survey Interview* (pp. 3–45), edited by Douglas W. Maynard, Hanneke Houtkoop-Steenstra, Johannes van der Zouwen, and Nora Cate Schaeffer. Hoboken, NJ: John Wiley & Sons. Copyright © 2002 by John Wiley & Sons, Inc. used by permission of John Wiley & Sons, Inc.

to be read verbatim in the order they appear" (Viterna and Maynard, 2002, p. 394), a practice that thus might be considered fundamental to standardization, and which appears even in Schober and Conrad's (1997) flexible or conversational interviewing. The version of standardization that Fowler and Mangione systematized is among the most rigorous and allows interviewers less autonomy than the practices used at most of the centers that Viterna and Maynard describe.

A comparison between the version of standardization articulated by Fowler and Mangione and the rules assembled by Brenner (1981a, pp. 19–22; 1981b, p. 129) illustrates how methods of standardization vary. Brenner allows interviewers to "show an interest in the answers given by the respondent," to "volunteer" clarification "when necessary," and to "obtain an adequate answer by means of nondirective probing,[3] repetition of the question or instruction, or nondirective clarification" when the respondent gives an inadequate answer. If a respondent asks for clarification, an interviewer must provide it, but nondirectively, by using "predetermined clarifications." The predetermined clarifications described by Brenner appear to be similar to the information that some survey centers routinely provide to interviewers in "question-by-question specifications" or that Schober and Conrad (1997; Conrad and Schober, 2000) provided to the interviewers in the experiments on "flexible" or "conversational" interviewing described later. Such practices contrast with Fowler and Mangione's (1990) recommendations for severely restraining the discretion of interviewers.

The variability in methods of standardization can also be illustrated by considering the practice of "verification," which some survey organizations, such as the Census Bureau, appear to consider compatible with Fowler and Mangione's first principle. Verification is a practice that acknowledges that respondents sometimes provide information before the interviewer asks for it, and that if an interviewer ignores the fact that she has already heard the answer to a question, the interaction may be awkward for both the interviewer and respondent. For example, an interviewer who has already been told the respondent's age in one context, instead of asking a later, scripted question about how old he or she is, might say, "I think you said earlier that you were 68 years old. Is that correct?" (Kovar and Royston, 1990, p. 246).

The second principle, to probe nondirectively, can also be implemented in different ways. For example, Fowler and Mangione (1990, pp. 39–40) require interviewers to repeat all the response categories when they probe closed questions. In support of this rule, there is research showing that the meaning of any individual category depends on the entire set of categories the respondent considers (Schaeffer and Charng, 1991; Smit et al., 1997). But there is variation in how this principle is implemented. For example, van der Zouwen and Dijkstra (2002) describe a practice called "tuning" in which the interviewer needs to repeat only the categories that appear to be in the vicinity of the respondent's answer. Tuning is itself similar to a technique called "zeroing in" that Fowler and Mangione (1990, p. 41) describe for numerical answers.

Survey practitioners recognize that there are situations in which the behavior of the interviewer cannot be standardized completely. For example, when investigators train

[3] A "directive probe," sometimes called a leading probe, is one that directly or indirectly offers a candidate answer; for example, "So would you say yes to that one?"

interviewers using "question-by-question specifications" of question objectives, the investigators recognize that the interviewer may need this information, the application of which cannot be prespecified, to diagnose and correct respondents' confusions and misunderstandings. Interviewers become less standardized when they actually apply the information provided to them during training, but that is part of the tension latent in traditional styles of standardization. Another example of a situation that cannot be highly standardized is probing, for example, of "Don't know" answers. Probing by the interviewer is associated with increased interviewer variance (Fowler and Mangione, 1990, p. 44; Mangione et al., 1992); that is, some of the variation in the answers is due to variation in the behavior of the interviewer rather than to variation in the concept being measured.

The rationale for standardization is that standardization decreases behaviors of interviewers that could give rise to variance in the answers that is not due to the concept being measured; reducing variation in the behavior of the interviewers should reduce error variance. In principle, when standardization is successful, it makes the effect of the interviewer in the aggregate very small, in the specific sense that it reduces the interviewers' contribution to variance. Standardization should also reduce bias by reducing the number of opportunities for the interviewer's expectations or opinions to intrude on the process by which the respondent's answer is generated, interpreted, or recorded. Studies in centralized telephone facilities where standardization is practiced, for example, find that the component of variance due to differences in the behavior of the interviewer across respondents (interviewer variance) is usually quite small for most types of survey items (Groves and Magilavy, 1986; Mangione et al., 1992). This is so even though there are many lapses in achieving the ideals of standardization (Oksenberg et al., 1991).

Standardization does not eliminate the effect of the individual interviewer on the individual respondent. An interviewer who follows the rules of standardization might react to a respondent's ambiguous answer by repeating all the response categories, for example. From the point of view of standardization, this is preferable to alternative behaviors such as the interviewer's using her beliefs about the respondent's likely answer to choose which response categories to repeat. But the standardized repetition of the response categories influences how respondents express their answers, and, over the course of the interview, probably has the desirable (from the point of view of facilitating the standardized interview) effect of training respondents to choose a category from among those offered. An answer is clearly an effect of the individual interviewer on the individual respondent. Nevertheless, if standardization were comprehensive and perfectly implemented, we could say that the effect of any interviewer would be the same as the effect of any other interviewer. That is, standardized interviewers are interchangeable.

This chapter does not examine the impact of standardizing the interviewer's behavior on the reliability or validity of the resulting data. Studies of interviewer variability are difficult to execute and are small in number as a result. A study comparing interviewer variability for standardized and unstandardized interviewing faces the challenge of selecting an appropriate comparison: Does one compare different varieties of standardized interviewers to each other or standardized to unstandardized interviewers, and, if the latter, how would those unstandardized interviewers be trained to behave, how would their answers be recorded, and how should outcome variables be specified? Sites within standardized interviews where standardization breaks down—for example, because of poor question design or poor training—provide a glimpse of the impact of standardization on reliability and validity. Collins (1980) described an instrument that omitted critical filter questions (questions that determine whether or not a target question should be asked); because of this design flaw, interviewers had to decide when target questions should be asked, and the result was a large component of interviewer variance in the answers. Interviewer effects were also present for the number of mentions to open questions and when probing was involved (see also Mangione et al., 1992). With respect to validity, a record-check study that compared respondents' reports about legal custody of their children to court records found that nonparadigmatic question-and-answer sequences in which the interviewer engaged in follow-up behaviors were less likely to yield accurate answers than sequences without follow-up by the interviewer (Schaeffer and Dykema, 2004). (However, we note that it is as plausible that a respondent's difficulty in answering occasioned the follow-up as that an interviewer's follow-up behavior somehow led to the inaccuracy.)

Because demands on interviewers—such as determining which questions to ask, asking open questions, probing, and defining complex concepts—are greater when interviewing is not standardized, many researchers are reluctant to experiment in loosening the rules of standardization. One such experiment is the event history calendar (EHC), and the initial experience reported by Dijkstra, Smit, and Ongena (forthcoming) illustrates the challenges of preparing interviewers for nonstandardized interviewing. In the versions of the EHC developed by Belli and colleagues (Belli, 1998; Belli et al., 2001; Belli and Callegaro, forthcoming), interviewers do not have to ask a series of questions in a particular order, and they are trained in procedures designed to improve recall of autobiographical events. When they evaluated a paper-and-pencil EHC administered by telephone versus a traditional standardized interview (using reports of events that occurred two years before the interview as a criterion) Belli, Lee, Stafford, and Chou (2004) did not find heightened levels of interviewer variance in the EHC; but preliminary analyses of life course reports gathered via CATI interviewing indicate that many items appear to have slightly higher interviewer variance in EHC interviews (Belli, personal communication, 2007). Even if EHC methods slightly increased interviewer variance, however, the total response error of EHC methods compared to traditional standardized interviewing could be smaller if EHC methods also reduced response bias.

It is plausible that the various components of standardization have different effects on the quality of data and that the size and direction of effects depend on the type of question (e.g., events, social categories, evaluations). Consider findings about the impact of reading the questions as worded. This practice appears to increase reliability, which is consistent with a reduction in the interviewer component of variance (Hess et al., 1999). And accurate reading of questions sometimes yields more accurate answers (Schaeffer and Dykema, 2004), but sometimes does not (Dykema et al., 1997). In a study that compared survey answers to court records to assess their accuracy, Dykema (2005) examined the impact of interaction on the accuracy of answers about characteristics of court orders about child support and legal custody (seven variables), the years of several salient events such as marriage and paternity establishment (four variables), and the amount of child support exchanged (one variable). Whether or not the interviewer made major changes in the wording of the question had no significant impact in nine cases and was associated with less accurate answers in two cases (whether the visitation arrangement was legalized and whether the parents had joint legal custody) and with more accurate answers in one case (amount of child support exchanged). When a cumulative pattern of behavior by the interviewers was assessed (errors in reading questions up to the question being evaluated), the pattern of interviewers' making major changes was associated with less accuracy for five of the seven questions about the court order and two of the four questions about years of events; as before, the pattern was reversed for the amount of child support exchanged. Such different results could occur because of variation in the difficulty of the task, because of problems in the design of the particular questions, or because of variation in the quality of the criterion.

Research suggests that other departures from standardization can also affect the quality of the resulting answers. For example, as already mentioned, when interviewers follow up by repeating only some response options or express their opinions, the response distribution of answers is affected (Smit et al., 1997), and when interviewers (who are standardized in that they are trained to read the question as worded) diagnose and correct comprehension problems and instruct respondents to ask for clarification, respondents give answers more consistent with survey concepts (Conrad and Schober, 2000; Schober and Conrad, 1997).

2.4 DATA

Up to this point, we have presented a description of the standardized interview in an idealized form. It is useful to temper this view by examining some examples from standardized interviews as they exist in practice. In the next sections we illustrate features of standardized survey interviews with excerpts from digitally recorded interviews from the 2004 wave of the Wisconsin Longitudinal Study (WLS). These observations were made in the course of developing a system for coding the interaction in the WLS interview. The health section, from which these excerpts are taken, comes early in the interview and has several types of survey item. The WLS began with a sample of 10,317 men and women, one-third of those who graduated from Wisconsin high schools in 1957; sample members have been interviewed several times in the intervening decades. The 2004 interviews were digitally recorded (with the permission of the respondents, who almost always consented), so that the interaction between interviewer and respondent is preserved along with the answers to questions. The WLS sample is structured in independent random replicates, or random subsamples. For our detailed examination of the interaction, we identified all cases in replicate two that were conducted by a single interviewer. From this group of cases we randomly selected 50 interviewers and sampled one interview from each interviewer, discarding cases in which the respondent did not give permission for the recording of their interview to be used in research. We made detailed transcriptions using conversation analytic conventions.[4]

2.5 PARADIGMATIC AND NONPARADIGMATIC QUESTION–ANSWER SEQUENCES

Because the question–answer sequence in which a question is asked as written, the respondent provides an answer in the format proposed by the question, and the interviewer acknowledges the answer is the same across interviewers, it could be considered the paradigmatic question–answer sequence of standardized interviewing (Schaeffer and Maynard, 1996). This paradigmatic sequence is illustrated by Excerpt 1.

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Excerpt 1:   WLS Case 10, Q38

FI:   # (0.2) .hh (0.1) A:n how would you duhscribe yer
      abi:lity tuh ruhmember things? (0.1) du:rrin' tha
      pa:ss fou:r wee↑:ks (0.1) .hh Were you a:ble tuh
      ruhmember mo:st thi↑:ngs (0.1) .h so:mewhat fergetful
      (0.3) ve:ry fergetful (0.4) or unable tuh ruhmember
      a:nythin' at a:ll.
      (0.3)
FR:   Mo:hss thi:ngs.
      (1.3)
FI:   °Nnka:y?°
```

In Excerpt 1 the interviewer reads the question as worded (line 1), the respondent identifies and clearly selects one of the offered response options ("most things," line 8), and the interviewer acknowledges that selection ("okay," line 10). This sequence presents the interviewer with a routine situation: she simply records the response option identified by the respondent. The paradigmatic sequence can take some embellishment, as in Excerpt 2, and still be recognizable:

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Excerpt 2: WLS Case 15, Q38

FI:  Oka:y? (0.1) .h Wood yuh duhscribe ye:r- (0.1) ho:w
     wood yuh duhscribe yer ability tuh reme:m↑ber
     thi:ngs durin' tha pa:ss four wee:ks? .hh Wuh you
     able tuh remember mo:st thi:ngs, somewhat fergetfu:l
     very fergetfu:l er u:nable tuh
     [remember a:nthin' at a:ll.]
MR:  [Yeh mo:st thi:ngs I pritty] much r'member
     °evrythin' pritty we:ll.° hh (0.4)
```

[4] Transcription conventions are described in the Appendix.

The response in Excerpt 2 is more complex than that in Excerpt 1: the initial "yeh" (line 7) is not a response to the survey question and cannot be coded by the interviewer, but it is followed by "most things," a phrase that appears in only one of the response options and can be heard as selecting that response option. The respondent then describes his situation as "I pretty much remember everything pretty well," a type of comment that we label a consideration. Even though "most things" is only a partial repetition of the response option and is preceded by an uncodable "yeh" and followed by a consideration, the skeleton of the paradigmatic sequence is recognizable, and the interviewer proceeds to the next question (not shown).

Paradigmatic sequences probably constitute a large majority of question–answer sequences (they do in the study examined here, for example), particularly if the questions are well designed and the interviewers well trained. The smoothness of the paradigmatic sequence does not in itself, however, indicate that either the question or the answer is of high quality. The respondent may interpret a question in a way different from the way the investigator intended without experiencing or giving evidence of any difficulty. Similarly, a paradigmatic answer may be a routine socially acceptable answer (e.g., "of course I do not cheat on my income taxes") or express an estimate as more certain than it is.

Thus, the fact that a sequence is paradigmatic does not necessarily indicate that a particular answer is valid or reliable, only that the respondent found a way to complete the task. Nor do deviations from the paradigmatic sequence necessarily indicate that data are of lower quality. It is easy to find examples in which a respondent's answer is not in the format requested by the survey question, the interviewer records the answer in the response categories without probing, and it appears that the quality of the data has not been compromised (Hak, 2002; Schaeffer and Maynard, 1996). [However, Schaeffer and Dykema (2004) found that in the case of legal custody, answers that interviewers treated as implicitly codable in this way were no more accurate than uncodable answers.] When the respondent asks for clarification, the question–answer sequence is no longer paradigmatic, but we have already noted that such requests sometimes are associated with more accurate answers (Schober and Conrad, 1997). Nevertheless, some deviations from the paradigmatic sequence, such as qualifications, are repeatedly associated with reduced accuracy (Draisma and Dijkstra, 2004; Dykema, 2005; Schaeffer and Dykema, 2004).

Paradigmatic sequences provide little information about the understandings, misunderstandings, thinking, accommodations, or adjustments that contributed to the respondent's answer. That is, paradigmatic sequences provide little information about the interpretive practices (or the cognitive process) by which they are constructed. Nonparadigmatic sequences can be informative in several ways. For example, nonparadigmatic sequences may display the respondent's understanding of the question, understanding of the task, level of uncertainty, facts the respondent is considering or rejecting, and so forth. These and other features of nonparadigmatic sequences have usefully been interpreted as evidence of problems in question design (Fowler, 1992; Oksenberg et al., 1991; van der Zouwen and Smit, 2004) or the quality of the resulting data (Draisma and Dijkstra, 2004; Dykema et al., 1997; Hess et al., 1999; Schaeffer and Dykema, 2004). Nonparadigmatic sequences also often constitute occasions on which the interviewer may—or must—react in some way that cannot be completely prescribed by the practices of standardization. These sequences thus provide sites to observe the participants deploying conversational practices learned in other social situations within the structure of the standardized interview. Actual practices reflect the "tacit knowledge" (Maynard and Schaeffer, 2000) that interviewers and respondents use to make the interview happen. One way in which the paradigmatic sequence breaks down occurs when the respondent fails to provide a properly formatted answer, and later we examine some of the ways in which this can happen and the interactional skills that participants use to conduct the task at hand.

2.6 CONVERSATIONAL PRACTICES WITHIN THE STANDARDIZED INTERVIEW

An early view of the survey interview characterized it as a "conversation with a purpose" (Bingham and Moore, 1924, cited in Cannell and Kahn, 1968), and this view of interviews as conversations was later echoed in the description of survey interviews as "conversations at random" (Converse and Schuman, 1974). In contrast to these informal characterizations of the survey interview stand the formal rules and constraints of standardization that we have just described. Someplace between a "conversation with a purpose" and a perfectly implemented standardized interview are the actual practices of interviewers and respondents.

Survey methodologists have studied the interaction between the interviewer and respondent for many years (Cannell et al., 1968). Most examinations of interaction in the survey interview have used standardization as a starting point and focused on how successfully standardization has been implemented, for example, by examining whether interviewers read questions as worded or respondents provide adequate answers (Brenner, 1981b), although exactly how these actions are defined varies. However, as researchers have looked more closely at what interviewers and respondents do, they have noticed and described in more detail how the participants import into the survey interview conversational practices from other contexts (Maynard and Schaeffer, 2006; Schaeffer, 1991; Schaeffer and Maynard, 2002; Schaeffer et al., 1993; Suchman and Jordan, 1990). As such observations have accumulated, they provide material for considering what these conversational practices accomplish, how they might support or undermine the goals of valid measurement within the survey interview, and how they might subvert or support the practices of standardization.

2.7 ACTIONS OF INTERVIEWERS AND RESPONDENTS

It is possible to examine nonparadigmatic sequences from a stance that attempts to characterize the utterances of interviewers and respondents and their sequential organization into actions. The actions of interviewers and respondents can be described at several levels. At the most general level, the participants ask and answer survey questions. At an atomic level, they deploy particles, talk, or remain silent. At intermediate

TABLE 2.1 Some Utterances and Actions of Interviewers

Utterances	Actions
Read survey item verbatim	Perform question from interview script
Modify survey item	Train respondent
Repeat survey item verbatim	Follow up to correct comprehension problems
Repeat response options verbatim	Follow up to correct inadequacies in the answer
Repeat answer	Motivate respondent
Propose formatted answer	Maintain interaction
Accept or code answer	Record answer
Signal return to script	Compliment
Request clarification	Reassure or encourage
Evaluate respondent's performance	Apologize
Say "sorry"	Affiliate
Laugh	
Comment	
Digress	

levels are utterances such as repeating questions, repeating answer categories, paraphrasing questions, and so on, and actions such as diagnosing problems or training respondents. One way of thinking about these utterances and actions is illustrated in Tables 2.1 and 2.2. Actions are accomplished as sequences of utterances, some of them prescribed by standardization, others improvised. Some of the utterances and actions that a description of interviewing must account for, such as reports and immediate coding of answers, have been described before (Hak, 2002; Schaeffer and Maynard, 1995, 2002), but others, such as encouragement and compliments, have received little attention (Gathman et al., 2006).

The lists of utterances and actions in Tables 2.1 and 2.2 are not comprehensive, and the distinction between utterances and actions is blurred in some cases. Nevertheless, these lists display some features of interaction that must be accounted for in

TABLE 2.2 Some Utterances and Actions of Respondents

Utterances	Actions
Repeat question or part of question	Interrupt
Repeat or paraphrase one or more response options	Display understanding
Use affirmative	Display retrieval of information
Use negative	Display consideration
Report with consideration	Display uncertainty or difficulty
Report with hesitation or other marker	Display construction of answer
Report with a quantification	Request clarification
Ask question	Provide answer
Laugh	Show a stance or position
Comment	
Digress	

a description of the survey interview. What some of these features look like, and the way that utterances build toward actions, can be shown with a few illustrations. In the following excerpt, the respondent interrupts the interviewer's initial reading of the response options:

Excerpt 3: Case 49, Q39

```
FI:  .hh A:n u:m- ho:w wood you duhscribe yer ability tuh
     thi:nk an so:lve- day tuh day pro:bl'ms, u:m durin'
     tha pa:ss four wee:ks, .hh a↑gai:n were you a:ble
     tuh think clearly 'n' solve probl'ms, hadduh little
     di:fficulty:, had [so:me-  ]
MR:                    [A:h clea]:rly.
FI:  .h Oka:y, I'm sorry I'm juss required tuh read
     a:llree- all thee o:ptions fuh these. .hh U:m had
     so:me di:fficulty:, (0.1) hadduh great deal of
     di:fficulty:, o~:r u:nable tuh thi:nk, °o~:r so:hlve
     pro:bl'ms.°
     (0.1)
MR:  Na::h. (0.2) (uh)
     (0.1)
FI:  Uh=whi:sh [o:ne,] >did you w'nt-<
MR:            [Put- ]
MR:  Put clea:rly.
FI:  Oka:y, tha firs' one? Thi[:n-]
MR:                           [Ye ]:ah.
FI:  Oka:y, (0.1)
```

In response, the interviewer's next utterance announces that she will read all the options, which she then does. The respondent's "nah" (line 13) is insufficient to identify which option he is choosing, and the interviewer's next utterance explicitly asks for that identification. The interviewer's utterances roughly meet the requirements of standardization outlined earlier. Because of their sequential placement, the interviewer's utterances constitute follow-ups or probes. At a more general level the probes are an interactional method of training the respondent that he or she must provide an answer that is formally as well as substantively adequate (Moore and Maynard, 2002).

In Excerpt 4, the interviewer's intervention comes before the respondent speaks:

Excerpt 4: Case 49, Q40

```
FI:   .hhh (0.1) A:nd have you ha:d any trouble with
      pai::n or disco:mfort durin' tha pa::ss fou:r
      wee:ks?
```

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 4          (1.0)
 5    MR:   U:::h hhhh
 6          (1.6)
 7    FI:   (hh)
 8          (0.8)
 9    FI:   An this is u:m- (0.5) completely up tuh ye:r
10          (incare-) interpreta:tion of i~t.=
11    MR:   =I guess ri:lly no:t. h
12          (.)
13    FI:   O¯ka:y yuh want me tuh put no: ↑fuh that izzat
14          righ'?=
15    MR:   =Ye:ah.=
16    FI:   =°Okay,°
```

In Excerpt 4, the interviewer treats the respondent's particle and the silences (lines 4–8) as potentially indicating a problem, and her utterance proposes what might be a simplification of the response task ("and this is um... up to your... interpretation of it," lines 9–10). The interviewer's utterance here is a preemptive probe (Schaeffer and Maynard, 1995, 2002), which responds to the delays in lines 4–8 by reducing the difficulty of the task.

2.8 REPORTS IN STANDARDIZED INTERVIEWING

One way in which the paradigmatic sequence breaks down occurs when the respondent provides a "report." In Drew's (1984) description of reports, the recipient of an invitation (e.g., "Do you want to have dinner on Sunday?") did not reply directly, but instead provided information relevant to a refusal of the invitation (e.g., "Isn't that the night of the opera?"), and left the upshot to the person issuing the invitation. In survey interviews, what we label a report is an utterance by the respondent that is provided in the position where an answer (to the original reading of a question or to a probe) is expected, that does not use the format projected by the question (e.g., the utterance is not "yes" or "no" to a "yes/no" question), but that provides information relevant to the process of answering the question (i.e., it is not a digression). Thus, the report provides an opportunity for the interviewer to gather or propose the upshot of the report and to record an answer or take some other action (see Moore, 2004; Schaeffer and Maynard, 1996, 2002).

This basic structure can be seen in the following excerpts:

Excerpt 5: WLS Case 10, Q2

```
1   FI:  # (0.3) .t .hh Du:rrin' tha pa:ss fou:r wee↑:ks
2        have you: been able tuh see well enu↑:ff tuh read
3        o:rdinary newsprin' .hh (0.1) withou:t gla:sses er
4        contac' le:nses?
         (1.2)
```

```
FR:  A:h- (0.1) Ju:ss rea:din' gla:sses?
     (0.7)
FI:  Oka:y? I'm 'ust gonna reread tha que:stion? .hh=
FR:  =O↑:h [o  ]ka:(h)y h (0.1) .hh=
FI:         [uh-]
     Du:rrin' tha past fou:r wee↑:ks (0.1) .hh have you
     been able tuh see well enu:ff (0.1) tuh read
     o:rdinary news↓print (0.1) .hh with↑ou:t (0.2)
     gla:↓ss~es (0.1) o:r co:ntac' le:ns~es.
     (0.3)
FR:  A:~h- (0.3) No↓:. hh
FI:  Oka:y?
```

Excerpt 6: WLS Case 1, Q7

```
FI:  Yer doin' be~tter than I a(h):(h)m. (0.1) .hh (0.1)
     Without a hearing aid an while in a group
     conversa:↑tion with at least three other pee↑pu:ll
     have you been able tuh hear what is sai:d?
     (1.7)
MR:  Not very goo:d. hh[h]
FI:                    [h]hh Would you say no:?
     (0.6)
MR:  No[:.]
FI:    [°iz]zat-° (0.1) .hh (A:hem) what about wi:th a
     hearing aid.
     (1.5)
MR:  U:m be:tter cha:nce the:n.
FI:  °Kay.° (0.2) Wou:ld you say yes er no in ge:nrull.
```

Excerpts 5 and 6 present examples in which the respondent does not provide a formatted answer but instead states relevant information without providing an upshot. In Excerpt 5 the interviewer responds to the report ("Ah. . . just reading glasses," at line 6) by announcing that she is going to reread the question (line 8) and then doing so (lines 10–14). Two additional examples of reports and the way interviewers respond to them are presented in Excerpt 6, and we begin with the second example in that excerpt. When answering the second question (lines 10–11) about whether he can hear with a hearing aid, the respondent provides the report that there is a "better chance then" (line 13). In line 14 the interviewer acknowledges the answer ("okay") and uses a standardized probing technique to ask the respondent to select a response category ("yes or no"). But she also seems to treat the report as indicating a trouble in answering; she downgrades her request with a phrase, "would you say," which we refer to as a distancing phrase, and simplifies the task ("in general"). In both cases, the utterance of the interviewer, in effect, returns to the respondent the task of producing the upshot of the report. As actions, the interviewers' utterances are follow-ups that support the practices of standardization by instructing the respondent to attend to details of the question when formulating an answer, but in neither of these cases does the interviewer suggest an answer.

In the first example of a report in Excerpt 6 (“not very good,” line 6), the interviewer displays another method that interviewers commonly deploy for responding to a report: proposing a candidate answer (“Would you say no?” line 7) in a directive probe. This sequence is similar to sequences that Moore and Maynard (2002) examine in which the answer is “formally” (which they contrast with “substantively”) inadequate. In their corpus of 3880 sequences from five surveys, Moore and Maynard (2002, p. 304) observe that a directive probe is much more likely when the respondent gives an answer that is formally inadequate than when the answer is substantively inadequate. As in this case, however, it is not always clear when the inadequacy of an answer should be considered “formal” rather than substantive, and the requirements of standardized measurement in this respect may be more constraining than those of other social contexts. If the inadequacy of the answer is truly simply formal, the quality of the resulting data is unlikely to be affected by this sort of directive probe; the difficulty lies in determining when the inadequacy is only formal. (We return to this example in our discussion of mitigators that quantify.)

The respondent in Excerpt 5 presents relevant facts or considerations in her report (“just reading glasses,” line 6), and we use the term “considerations” to describe this type of report. These and other types of reports (described later) show respondents taking a step toward fitting their experience into the terms of the survey question and failing to do so. Reports present occasions on which measurement error is revealed to be present (e.g., by the respondent’s uncertainty) or perhaps introduced (e.g., by the interviewer’s response). In addition, our examples show interviewers using their tacit knowledge to alternate between the formal practices of standardized interviewing and conversational practices as they attempt to negotiate a complete formatted answer (Maynard and Schaeffer, 2000).

2.9 REPORTS THAT MITIGATE: UNCERTAINTY AND GENERALIZATION

In addition to presenting considerations, reports can also display doubt or uncertainty:

Excerpt 7: WLS Case 30, Q12

```
 1   MI:   .hh (0.1) A:n- (0.1) hev pee↑pull who do not kno:w
 2         you: understood you completely when you spea:k?
 3         (0.6)
 4   MR:   Ah thi:nk so:. (0.1) [#] (0.1)
 5         I still gotta little bit of uh
 6         w- Wis[co]:nsin accent, (0.1) but [(these fo:lks, ]
 7   MI:         [# ]                        [HH hu(h) hu(h)h]
 8         (0.2)
```

```
 9   MI:  HU~ (H)H [.HH ]
10   MR:            [They] j'ss put that asi:de,
11        (0.1)
12   MI:  O(h)ka(h):y, (0.2) U:h- wood you say like- (0.2)
13        ye:h on that o:ne (0.1) iz[zah c'rreh?]
14   MR:                            [Ye(h):    ](h)h.
15        (0.2)
16   MI:  O(h)ke(h):(h)h. .hh
```

In Excerpt 7, the respondent provides a conventional expression of uncertainty ("I think so," line 4) and follows that with a self-deprecating joke or humorous offering (lines 5–6, 10). The answer is not properly formatted and leaves the upshot to the interviewer. The interviewer treats the respondent's answer as requiring only confirmation and produces a directive probe (lines 12–13). This type of report (like others, such as giving ranges) embodies practices that participants engage in to express uncertainty or otherwise mitigate answers, and interviewers commonly confront such answers as they conduct standardized interviews.

Another type of report represents a practice that may be more common when answering questions about threatening topics, that is, questions that ask for information that might be embarrassing or incriminating (Schaeffer, 2000):

Excerpt 8: WLS Case 4, Q35

```
 1   MI:  Oka:y, (0.1) A:n durin' tha pa:st four wee:ks,
 2        didjuh ever fee:l fre:tfu:l, a:ngry, irrituhbull
 3        ankshuss er duhpressed?
 4        (0.7)
 5   MR:  O:ah I think evrybody gets i:rrituhbull,
 6        (0.3)
 7   MI:  °Mm↑hmm° So: I should sa::y- ye:s t'that
 8        que:ss[ch'n?]
 9   MR:        [Ye   ]:ss.=
10   MI:  =O:ka:y, (0.2) .hh (0.1)
```

In answering the question about his recent emotional experiences, the respondent does not describe his own behavior. Instead, the report provides a speculative generalization (line 5), which, by referring to "everybody," can be heard as including the respondent without his having to make a direct admission. As with other reports, this one provides an opportunity for the interviewer to gather or propose an upshot, which he does with a directive probe that is crafted as a request for confirmation (lines 7–8). In this case, the interviewer's directive probe can be heard as relieving the respondent of the obligation to initiate a potentially embarrassing admission about himself.

2.10 REPORTS THAT QUANTIFY

In the next example, the respondent, rather than using the dichotomous response options, produces an answer that suggests a scale or continuum:

Excerpt 9: WLS Case 28, Q32

```
FI:  HH=O(h):ka:y? .HH Durin' tha pa:st four ↑wee:ks .hh
     have you been feelin' ha:p↑py or u:nhappy.
     (3.1)
MR:  O:↑::hhh I guess I'm- (0.2) feelin' pritty ha:ppy.
     (0.5)
FI:  A::ri:ght.
```

Offered a choice between happy and unhappy, the respondent in Excerpt 9 displays a stance of uncertainty (with the mitigating phrase, "I guess," line 4), incorporates a word from the question ("feeling"), and volunteers a response that was not offered as a category ("pretty happy"). By using a modifier ("pretty"), the respondent displays an orientation to a scale with gradations of happiness. An early scaling study indicated that "pretty" happy is less happy than "happy" (Cliff, 1959), and conversation analytic research suggests that a phrase such as "pretty good" can be a "qualified response to a how-are-you inquiry" (Schegloff, 1986, p. 130). Because survey measurement requires classification in the offered categories, "pretty happy" presents the problem that it is not clear whether it should be classified as "happy" or "unhappy." The interviewer solves this difficulty by classifying the respondent as "happy" (this can be determined by seeing which question is asked next), thus engaging in a practice that has been called immediate coding (Hak, 2002), without confirming her action with the respondent. The conversational practice of treating the upshot of answers such as "pretty happy" may be so commonsensical and routine that the interviewer may not notice that she recorded the answer in a form different from that in which the respondent expressed it.

In the next two excerpts, the respondents' answers may display that they are experiencing difficulty in translating a value into the two implicit response options, yes and no. The question in Excerpt 10 and Excerpt 11 (which reproduces part of Excerpt 6) asks the respondents to classify themselves with respect to several conditions: without a hearing aid, in a group conversation, with at least three other people, and able to hear what is said. For the last condition, in particular, it may not be clear how to determine when the condition is satisfied:

Excerpt 10: WLS Case 28, Q7

```
FI:  A::↑ri:gh(t) .hh Wwithou:t a hea:rin' ai:d, an while
     in grou:p conversa:↑tion with at least three other
     pee↑pu:ll .hh have you been able tuh hea:r what is
     sai:d?
     (1.8)
```

```
 6   MR:  Mo:stuh tha ti:me.
 7        (1.2)
 8   FI:  I:zzat a ↑ye:s zen e[:r-]
 9   MR:                      [Ye ]:s
10        (0.1)
11   FI:  O[ka:y?]
```

Excerpt 11: WLS Case 1, Q7

```
1  FI:  Without a hearing aid an while in a group
2       conversa:↑tion with at least three other pee↑pu:ll
3       have you been able tuh hear what is sai:d?
4             (1.7)
5  MR:  Not very goo:d. hh[h]
6  FI:                    [h]hh Would you say no:?
7             (0.6)
8  MR:  No[:.]
```

Each of these reports displays an orientation to a continuous dimension: the first focuses on a frequency dimension (how often), the second on an intensity dimension (how well). It is not clear whether being "able to hear what is said" means that the respondent is able to hear what is said "ever" or "always" or with some other frequency. Similarly, being able to hear what is said could mean that one is able to hear "even a little bit" or "extremely well" or somewhere in between. The question wording provides no guidance to the respondent or the interviewer in deciding where the threshold for saying "yes" lies. The reports of these respondents, like that of the respondent in Excerpt 9, implicitly poses to the interviewer the question of where on the continuum lies the boundary between the offered response options, and each interviewer responds with a directive probe.

In the following excerpt, a similarly complex question that nevertheless asks for a yes or no answer is also met with a response that refers to a continuum, although in a humorous way:

Excerpt 12: Case 29, Q17

```
 1  MI:  Have you been able to be:nd (0.1) li:ft
 2       (0.1) ju:mp a:nd ru↑:n without di:fficul↑ty:
 3       (0.1) an without he:lp or equip↑ment of any
 4       ki:nd? .hhh
 5             (1.2)
 6  FR:  A:↓:h no:t too much ju:mpin'
 7       he[(h)hhe(h)h]he(h)hhe[(h)h↓he(h)h]
 8  MI:    [O:kay?    ]        [so the     ]:n I shud
 9       put no:the:n fuh tha:t one [is that c'r↑reh-]
10   FR:                            [A:h tha:t's     ]
11       c'rre:ct ye:s.
```

The answer “not too much jumping” (line 6) addresses only one of the conditions mentioned in the question, and it does so in a way that may suggest that there is “no jumping,” “a little” jumping, or possibly the ability to jump but not the occasion to do so. In addition, the respondent appends laughter to her response (line 7), thereby displaying “troubles resistance,” and invites the interviewer to laugh in response (Jefferson, 1984; Lavin and Maynard, 2001). The interviewer delivers a directive probe, declining the invitation to laugh, ignoring many possible lines of inquiry, and moving the interview on to the next question.

2.11 DISCUSSION

In this chapter we introduced the standardized interview, the paradigmatic question–answer sequence, and the prescriptions of standardization that guide interviewers in handling routine and unexpected situations. We have also briefly described some of the ways that standardization can break down when respondents import common conversational practices into the standardized interview. In addition to reports, which we have examined in detail here, such practices include producing qualified responses to questions that ask for a response option, joking and laughing, and others (Lavin and Maynard, 2001; Schaeffer and Maynard, 1995, 2002). Faced with reports, interviewers may use canonical practices of standardized interviewing such as rereading the survey question. Or, and perhaps more commonly, they may routinely use their tacit knowledge of conversational practices, for example, to interpret reports as codable or not.

The reports that we have examined use the slot after the survey question to display a number of things. The considerations recited by respondents appear to describe aspects of their experience that they are using to construct an answer or that pose some problem as they attempt to find an appropriate response option. Reports display other translation problems as well, as when the respondent elects a response option that was not offered or returns to the interviewer the task of deciding where on an underlying continuum the threshold for “yes” lies. Finally, reports display answers that are mitigated in some way (e.g., by uncertainty) although interviewers routinely treat these answers as codable. One characterization of reports is that they display a lack of fit between the survey question and the experience that the respondent is describing. The lack of fit may originate in uncertainty about the facts (state uncertainty) or difficulty determining which of the offered options represents the respondent’s true value (task uncertainty) (Schaeffer and Thomson, 1992), or somewhere else. Evidence has accumulated that some features of interaction—like reports, delays in answering, or qualification in answering—indicate measurement error (unreliability or invalidity) or problems in question design (Draisma and Dijkstra, 2004; Dykema et al., 1997; Hess et al., 1999; Mathiowetz, 1998, 1999; Schaeffer and Dykema, 2004; van der Zouwen and Smit 2004).

The examples that we have presented used conventional methods of interviewing: a human interviewer talking with a respondent over the telephone. The survey interview of the future is likely to increasingly rely on modes that turn tasks previously performed by interviewers over to a computer and that are self-administered (such as computer-assisted self-interviewing, CASI, or web surveys). Because self-administered instruments eliminate the interviewer component of variance (the part of unreliability that is due to variation in the behavior of the interviewer), the data they obtain are potentially more reliable than data an interviewer can acquire (O'Muircheartaigh, 1991). In addition, data from self-administered instruments may sometimes be more valid, for example, for threatening questions (Tourangeau and Smith, 1998). However, simple technological enhancements alone are unlikely to lead to substantial improvements in the accuracy of data; for example, adding audio to computer-assisted self-interviewing (audio-CASI) may not improve the validity of data more than CASI alone (Couper et al., 2003). Moreover, in the absence of an interviewer, when respondents encounter problems in answering survey questions, those problems may be more likely to lead to item nonresponse or abandoning the task altogether.

Future enhancements to self-administered instruments are likely to be of several types: the development of animated agents who might conduct the interview, the addition of some of the diagnostic skills of an interviewer to a CASI instrument, and the ability to accept spoken input from the respondent. Although self-administered applications in which the machine presents a human agent visually would still be considered relatively exotic, video interviewers have been used in some survey experiments (Krysan and Couper, 2003), and animated agents are increasingly feasible. The literature on interviewer effects suggests that the characteristics of animated agents or voices could have an impact on the data if those characteristics were relevant to the topic of inquiry (e.g., "race" of the agent might affect answers if the topic were racial attitudes). However, the physical features of an agent could provide investigators the ability to control interviewer characteristics (e.g., race of "interviewer" could easily be crossed with race of respondent), hold them constant, or randomize them. The copresence of even an animated agent has the potential to be experienced as motivating by the respondent; one effect could be to reduce item nonresponse, for example. Like other SAQs, such an animated agent would be almost perfectly standardized; that is, it would expose all respondents to the same stimulus, increasing the reliability of data compared to interviewer-administered instruments. In addition, an animated agent could slow the pace of the interview, which could improve comprehension and provide more time for retrieval. However, experience with audio-CASI, during which many respondents turn off the audio if that option is available, suggests that enforcing a slow pace could irritate the respondent.

Increasing the validity of data, however, could require using other methods to determine the requirements of enhanced automated interviewing and the "skills" that it would require—with or without an animated agent. Ethnomethodology (EM) and conversation analysis (CA), whose substantive inquiries involve embodied forms of practical knowledge and talk-in-interaction, represent tools for understanding human–machine interaction.[5] Our partial inventory of utterances and actions in the interview

[5] For a review of EM and CA see Maynard and Clayman (1991); for a recent overview of studies concerning technology and work see Heath and Button (2002); and for examples of EM and CA studies of human–machine interaction see Suchman (1987), Maynard and Schaeffer (2000), Heath and Luff (2002), and Whalen et al. (2002).

and our extended discussion of reports illustrates the sorts of phenomena that ethnomethodology and conversation analysis can identify, details that are informative in themselves and that also serve as the starting point for other types of inquiry (such as systematic coding and quantitative analysis).

We have not attempted to describe the full range of utterances and actions that occur within the survey interview. But we have used reports to illustrate a type of information that respondents provide in interviews that is not generally encouraged or captured by self-administered instruments. Such information is potentially informative to survey methodologists, for reasons already described, and also to analysts. Although research has not yet identified what meta-information (like that which the respondent provides in a report) is most useful to preserve, or how to incorporate it in analytic models to control for the effects of measurement error, the studies that have been done (some listed earlier) show the promise of such investigations. A challenge faced by an automated interviewing system would be to correctly interpret such reports—for example, that the respondent was expressing some doubt but was clearly leaning in one direction—and to then fashion an appropriate response—for example, to code "probably" as "yes." Automated interviewing systems have faced such challenges, for example, by treating a pause before entering an answer as a sign of trouble and spontaneously offering help or deciding when the length of a pause before answering is a sign of a problem in comprehension (Ehlen et al., 2005; Schober and Bloom, 2004; Schober, et al., 2000). Other chapters in this volume illustrate that the success of automated interviewing and other innovations in the future will depend in part on how well we describe and understand the person-to-person interviews of the present.

APPENDIX: TRANSCRIBING CONVENTIONS[6]

1. Overlapping speech
`A: Oh you do? R[eally ]`
`B:              [Um hmmm]`
	`Left-hand brackets mark a point of overlap, while right-hand brackets indicate where overlapping talk ends.`

2. Silences
`A: I'm not use ta that.`
`(1.4)`
`B: Yeah me neither.`
	`Numbers in parentheses indicate elapsed time in tenths of seconds.`

3. Missing speech
`A: Are they?`
`B: Yes because ...`
	`Ellipses indicate where part of an utterance is left out of the transcript.`

4. Sound stretching `B: I did oka::y.`	`Colon(s) indicate the prior sound is prolonged. More colons, more stretching.`
5. Volume `A: That's where I REALLY want to go.`	`Capital letters indicate increased volume.`

6. Emphasis A: I do <u>not</u> want it.	Underline indicates increased emphasis.
7. Breathing A: You didn't have to worry about having the .hh hhh curtains closed.	The ''h" indicates audible breathing. The more ''h's" the longer the breath. A period placed before it indicates inbreath; no period indicates outbreath.
8. Laugh tokens A: Tha(h)t was really neat.	The ''h" within a word or sound indicates explosive aspirations; e.g., laughter, breathlessness.
9. Explanatory material A: Well ((cough)) I don't know	Materials in double parentheses indicate audible phenomena other than actual verbalization.
10. Candidate hearing B: (Is that right?) ( )	Materials in single parentheses indicate that transcribers were not sure about spoken words. If no words are in parentheses, the talk was indecipherable.
11. Intonation A: It was unbelievable. I ↑ had a three point six? I ↓ think. B: You did.	A period indicates fall in tone, a comma indicates continuing intonation, a question mark indicates increased tone. Up arrows (↑) or down arrows (↓) indicate marked rising and falling shifts in intonation immediately prior to the rise or fall.
12. Sound cut off A: This- this is true	Dashes indicate an abrupt cutoff of sound.
13. Soft volume A: °Yes.° That's true.	Material between degree signs is spoken more quietly than surrounding talk.
14. Latching A: I am absolutely sure.= B: =You are.	Equal signs indicate where there is no gap or interval between adjacent utterances.
A: This is one thing [that I= B: [Yes? A: =really want to do.	Equal signs also link different parts of a speaker's utterance when that utterance carries over to another transcript line.
15. Speech pacing A: What is it? B: >I ain't tellin< you	Part of an utterance delivered at a pace faster than surrounding talk is enclosed between ''greater than'' and ''less than'' signs.

[6] Adapted from Gail Jefferson (1974), Error correction as an interactional resource. *Language in Society*, 2, 181–199.

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